



Google Maps can warn you

Google Maps can warn you if your ride goes off-route as a precaution against possible predators. <https://dgit.in/july19-80>



Instagram Explore tab gets ads

Instagram's explore tab will now show you ads after you've picked a story and scroll down the feed. <https://dgit.in/july19-81>

government reports, public data like census and other such sources. Once you have collected your data, the next step is efficiently cleaning this data. Data, unfortunately, doesn't usually get generated in an organised manner. It comes as an amorphous heap of data which needs to be cleaned, sorted, and filtered. The third step is arranging the data in such a way that you can easily analyse or make sense of it. Roy states, "Spreadsheets are your friend, as data journalists. Use them copiously and unabashedly". When the data is displayed in an organised manner, data journalists can deduce patterns from them, which ultimately confirms or refutes their hypothesis.

Roy also divulged that data is increasingly common in the form of extremely large data sets which can go up to a couple of GB sometimes. In such cases, other methods like non-relational databases are used. Data stories don't only require statistical and numerical data, but they also involve textual data which is where technologies like OCR and the usage of coding languages come in. After these processes, you must visualise your data by creating charts, graphs, and maps (heatmaps, interactive maps and more). This can be done using apps and software like Tableau, Charted, Datawrapper, Carto, D3, QGIS and of course, coding expertise (if you have it). This visualisation will help you understand the data and additionally, adds a nice visual touch to your stories as well. After all of these steps, the data journalist will finally begin writing the story. This story must not just be about the data, but must actually use a narrative technique to relay the story in a manner that can be understood easily, even by novices in the field you are covering. Simply put, the process is - collect data > clean and filter data > analyse data > visualisation of data > write the story.

CHALLENGES

Access to open, public data is an absolutely crucial element of data journalism. The mere availability of data serves no purpose to journalists. It must be easily and quickly accessible as well.

The Indian government usually provide open data through the RTI (Right to Information) Act. However, obtaining this comprehensive government data can take inordinately long. Thus, the time proves costly and the story never actually sees the light of the day. Another obstacle investigative data journalists face is that the government sometimes manipulates data. Refusals to answer RTIs are also at a record-breaking high in India. Roy states, "A lot of data is also getting locked up in the hands of private organisations. For example - Aadhar data. We don't have any legislative or judicial approach to force these organi-



Tableau is a great tool to create data visualisations on

sations to share public data. People must attempt to reclaim their data, just like they are reclaiming public spaces". Data journalists in India, however, have access to hordes of government-generated data through data.gov.in, which is a step in the right direction.

Another challenge aspiring investigative data journalists may face is the more complex learning curve that this profession entails in comparison to traditional journalism. Data journalists are required to have strong data analytical skills and versatile domain knowledge. It is also difficult to get data in easier formats (comma separated views, text format, spreadsheets, etc.) and a lot of effort goes into preparing data from particularly difficult sources. Ramani divulges yet another problem, "In our media environment, there is too much emphasis put on short-term and

'breaking news' stories. Investigations require time and effort. And fighting for your stories against the breaking news stories is difficult, especially if senior editors are unaware of the effort that is required to pull off data-based investigations". Additionally, although there are free courses online for data journalism, physical courses in India are very limited in this field.

SCOPE

We see many Indian media outlets delving into data journalism, especially investigative data journalism now. Notable outlets like The Hindu, Hindustan Times and more are covering in-depth investigative data stories that go into extreme detail of the topic they are reporting. Ramani is optimistic about the scope of this profession in India, he says, "In India, there are several large data sets on various aspects of economy, society, state and so on and journalists can use them to supplement their reporting." Young journalists that are born into today's digitally-enriched world will be more adept at picking up data preparation and digital visualisation skills.

Indian students, in some universities - such as Asian College of Journalism, Chennai and Symbiosis Institute of Media and Communication, Pune - are taught a module on data journalism. They are introduced to the latest, free tools that help with visualisation like Google Fusion Tables, Tableau Public, Carto and Datawrapper. Some, even introduce them to basic coding. In addition to this, there are also several open-source, free online courses that allow individuals to pick up rudimentary data journalist skills. However, Ramani believes that the best way to learn is to practice investigative reporting using data and by picking up skills as you go about doing your work. Investigative data journalism covers several domains, right from reporting sales data in business journalism to covering crime and corruption. Indian media needs to acknowledge that investigative data journalism can provide stories rich in content, analysis, figures and ultimately, even narrative. 